

Project 3

TITLE: Cloud Connected Security Node

OBJECTIVE:

To build a security monitoring system that sends distance telemetry data from ESP32 to a cloud dashboard.

COMPONENTS USED:

- * ESP32
- * HC-SR04 Ultrasonic Sensor
- * Adafruit IO Dashboard
- * Wokwi Simulator

SOFTWARE USED:

- * Arduino IDE
- * Adafruit IO
- * Wokwi Simulator

WORKING PRINCIPLE:

The HC-SR04 sensor measures object distance. ESP32 processes the sensor data and transmits it to Adafruit IO using Wi-Fi. The dashboard displays real-time distance readings for remote monitoring.

FEATURES:

- * Distance monitoring
- * Wi-Fi connectivity
- * Cloud dashboard integration
- * Intruder detection
- * Remote telemetry

RESULT:

The system successfully transmitted real-time distance data to the Adafruit IO dashboard.

CONCLUSION:

This project provided hands-on experience with cloud connectivity, IoT telemetry, and remote monitoring systems.